Checking or Recomputing? Two Cost Regimes in Verifying AI-Generated Formal Artifacts

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Abstract

Checking a stored proof can avoid its original search, while checking a computational certificate can repeat the calculation that justifies it. We measure these costs in a pinned, published AI-generated Lean library and use the findings to design ProofServe, an original-kernel check and audit tool. Matched target windows retain 80.28–82.35% of source-compilation wall time for two computational certificates, versus 0.46–6.54% for two ordinary controls. Across 31 module-process pairs, the median checker/compiler CPU ratio is 0.451, with overlapping source-proxy classes. A dependency-ordered measured prefix covers 128,331 of 163,096 modules (78.7%): 134.7 P-core and 256.7 E-core CPU-hours are separate observed-work lower bounds, not a full-library estimate. The missing tail is not imputed. The full-frame import graph omits 19.78% of source files and 33.30% of lexical kernel-decision sites. ProofServe combines dependency-aware parallel replay, restartable segments and receipts recording scope, axioms and costs. Its contribution is a usable integration of established mechanisms and scoped evaluation. Final release scaling and external-project measurements remain pending.

1 Introduction

Proof-embedded computation can preserve expensive work across the boundary between producing and checking an AI-generated proof. Lean validates elaborated terms rather than repeating tactic search, but definitional equality can still evaluate a program inside a term. Users therefore need both a cost model and a precise account of what a successful project check covers.

Existing work already establishes the value of independent checking. Lean 4.35 ships `lake check`, following the FRO's external-checking agenda [8, 7]. A public report on the same OpenAI library performs a complete cloud build and extensive independent checking [10]. It proposes kernel computation as an explanation for expensive checks from three source examples. Our empirical increment is a matched module-level measurement of that explanation, an observed dependency prefix with explicit condition-dependent lower bounds, and a coverage analysis at a different corpus revision. Our systems contribution is ProofServe's original-kernel workflow and auditable receipts, evaluated through its release-form interface. The four questions follow these boundaries:

RQ1 How do checking costs differ between stored mathematical proofs and computational certificates?

RQ2 What work is observed in the measured dependency prefix?

RQ3 How does parallel original-kernel checking trade wall time, CPU work and additional trusted code?

RQ4 What do default builds, repeated verdicts and axiom receipts establish about verification coverage?

2 Method and workload

A per-module observed ledger and matched checking windows separate measured library work from selected-case checking behavior. Our primary corpus is the published OpenAI mathematics repository [11], pinned at `4eaee875`: 163,096 executable modules plus one configuration unit, Lean 4.34.1 and Mathlib `d13f23b7`. Two original draws in each of 21 strata combine source-size percentile, import depth and an explicit kernel-decision marker. All 42 now have completed compilation costs. Prerequisites and selected hot examples retain their separate identities and weights.

Source cost includes a module's imports, tactics, kernel checking and artifact emission. Separate prerequisite preparation, C compilation and build-driver work are outside this clock. CPU means user plus system time. Whole-process replay and incremental target windows are distinct measurements; Figure 1 labels both. Reported sample and release timings use CPUs 0–7 of an i9-14900K, one task at a time, retaining file caches. Lean runs omit `-M`, including `-M0`; external memory protection is recorded separately. Earlier in-process memory checks and missing prototype persistence marking were our own method defects.

The public package `data/openai-math-4eaee875/` records sample identities, raw-cost provenance, clocks and estimates. This paper pins its original-sample summary to `1cc14067`. Historical replay measurements explain cost boundaries; system performance must come from committed release-tool results in `checker/results/` and `data/external-projects/`.

Compilation used eight P and fourteen E cores with 22-slot admission, memory-reserve drains and exclusive retries. The user stopped after the first segment; the proposed second admission policy did not run. The total-instruction gate failed: median 1.33% and P95 3.15% exceeded unchanged 1%/3% criteria. Cross-module comparison uses user-mode instructions, chosen after those results. P/E CPU is unconverted.

3 Two costs and observed library work

3.1 Computational certificates retain work

Matched target windows show expensive replay for computational certificates and cheap replay for ordinary controls. In Figure 1(b), Grid and Amplitudes retain 80.28% and 82.35% of their source compiler's wall time. Their replay components are 255.489 and 110.026 seconds; almost all of each is inside the kernel replay call. Mathlib Nat.Basic and NearPin retain 6.54% and 0.46%. Each target component reads, maps and checks that target after dependency preparation; each denominator is that module's compiler process. These shared boundaries quantify the expensive kernel replay anticipated by the report's computation hypothesis in the selected cases [10].

The mechanism is visible in the proof boundary. A tactic can search for a mathematical derivation and emit a term that records its successful steps. Replaying that term validates those steps without reconstructing the search. A finite certificate justified through `of_decide_eq_true` instead asks the kernel to validate the decision procedure's result through reduction. The compact wrapper can therefore retain evaluation over a large finite object. Our target measurement holds the declared input and original checking operation fixed and locates that retained cost in the replay window. Its practical consequence is that source length and a cheap closure average are weak planning signals for an individual certificate.

The broader paired panel shows variation rather than a clean binary split. Across 31 original modules, checker/compiler CPU has median 0.451: 0.391 for the 14 unmarked and 0.498 for the 17 marked modules. An explicit comment/string-masked `decide +kernel` site is a reproducible proxy for computation, while ordinary `decide` can also encode substantial evaluation [2, 5]. For example, the unmarked Ramsey K10 certificate uses 356.44 checker CPU-seconds against 1,725.13 source CPU-seconds (20.66%). A 20-module sample of the retained 190-module Ramsey prefix uses 35.91% in aggregate. That prefix covers 190 of 444 targets and is dependency-ordered; its sample describes the retained prefix.

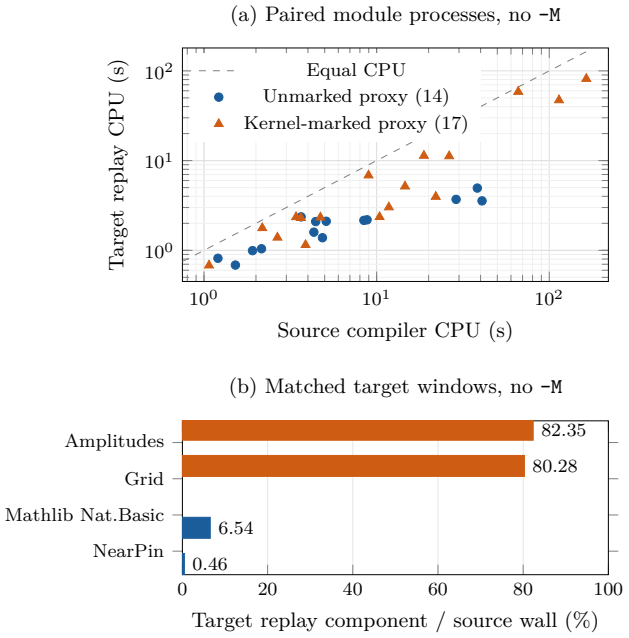


Figure 1: Two views of checking cost, all without `-M`. (a) Thirty-one complete module-process pairs; colors are source-marker proxies, whose distributions overlap. Imports are trusted in target replay. (b) Ordinary controls (blue) and computational certificates (orange), using matched target read/map/replay wall windows after dependency preparation. CPU in (a) and wall time in (b) have separate denominators. Design implication: expose the checking stage and amortize preparation; a closure average can hide an expensive certificate.

Whole closures dilute expensive targets with cheaper dependencies. Checkpointed Falconer checking costs 2,038.44/32,292.60 CPU-seconds (6.31%); matched checkpointed Mathlib costs 91.15/1,018.22 (8.95%). Their separate wall ratios are 9.26% and 14.67%. SubexponentialSAT primary accounting reports 3,399.30 checker CPU-seconds for all 14,721 accepted modules, or 7.25% of complete source work. Using only the 3,977 new targets' source cost gives 15.79% for the same whole-closure checker numerator. Its independent reference controls remain pending. This denominator change illustrates why a single ratio cannot stand for every boundary.

3.2 Observed compilation work and its limits

We measured 128,331 of 163,096 executable modules (78.7%); 34,765 remain unmeasured. Observed successful work is 134.7 P-core CPU-hours and 256.7 E-core CPU-hours, separate lower bounds rather than a full-library cost estimate. The executed interval is 24.62 wall-hours, including drains and retries.

The successful set finishes in dependency order, not

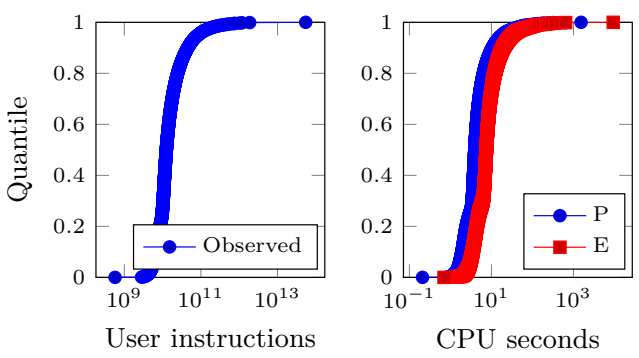


Figure 2: Empirical distributions for the 128,331 measured modules only: user-mode instructions, selected after the failed total-instruction gate, and separate P/E CPU. Dependency order and the unmeasured tail prevent population extrapolation. Interrupted attempts supply no successful cost; unmeasured modules are not zero-filled.

by random sampling. Unmeasured modules occur later in dependencies and include memory-intensive families; their mean cost may be higher. The observed work is a lower bound, not a full-library cost answer. Neither the original 42 samples nor a model fills missing costs. Every record retains source/measurement commits, pool, native CPU/wall and instructions. The one configuration unit has no native cost. There are no final failed modules, but 65 resource-interrupted attempts and one transition interruption remain separate from successful costs. The wall boundary is 4 November 19:52:10 to 5 November 20:29:39 UTC; summed module wall is a different clock.

All distributions and cost shares below describe only observed modules. Among them, 5,646 marked modules contribute 20.49% of observed P CPU and 20.95% of E CPU; 202 marked sources are unmeasured. The 25,828 observed modules outside the default graph contribute 20.16%/19.59% of P/E work. These source proxies do not identify kernel CPU phases or whole-frame shares.

External cost comparison. The earlier tooling report builds 122,458 modules at `fd4aeeb2` on 32 cloud vCPUs in 13.8 wall-hours [10]. Its approximate 410 CPU-hours comprise roughly 330 hours of compilation and 80 hours of Comparator checking. Compilation CPU is extrapolated from a quiet 26-module laptop sample, with estimated ±30% uncertainty. That estimate is an external comparison, not an observed per-module CPU ledger on our revision. The report's 26-module kernel-skipping ablation and roughly 30,900 cloud profiles remain build-side phase precedents. Our observed prefix retains its different revision, incomplete scope, hardware and post-hoc criterion; no calibrated agreement is presumed.

4 ProofServe design

ProofServe turns the cost and coverage findings into separate `check`, `audit` and compatibility-only `serve` interfaces. Its original-kernel route preserves mathematical checking rules while adding explicit coordination and artifact trust.

4.1 Check and resume a dependency closure

A checked prefix amortizes preparation while preserving each module's declared dependency context. `proofserve check <project>` discovers compiled local roots or uses explicit roots, traverses imports, and reports missing artifacts. The checker builds a graph from ordinary declarations' referenced constants. Ready declarations run against a checked frontier using the original kernel operation; the coordinator validates returned identities and commits the next frontier after all workers in a wave join. Nonordinary declarations and their ordinary prerequisites use the original replay path before the remaining graph runs. Unsupported local structure retains conservative fallback.

Review-driven repairs and theorem-body policy. Two read-only reviews of `e0c92d51` and `2628d727` traced coordinator defects; implementation-line regressions reproduced shadowed inputs, omitted axioms, unsafe duplicates, proof-body context substitution and malformed Quot restoration. Earlier O execution controls also exposed cross-project certificate reuse and added bad-proof parts. Repairs bind inputs, validate original contexts and quotient tags, separate worker events and preserve nonzero-exit dominance. The proposed project-IR stdout-forgery route remains unconfirmed. The issues have regression repairs; the repaired binary has no independent runtime review.

The frozen checker is `e000ed62`, replacing `e1557c77`. Its default *shared-checked* policy retains a kernel-checked theorem body when a later theorem has the same interface. A crafted serialized proof that passes only by depending on another module's theorem body can therefore be accepted while official checking rejects it in its declared import context. The accepted proposition is still derived from kernel-checked proofs in the shared checked closure and its reported axioms; this is a different artifact-context guarantee, not original-proof replay parity.

For matching theorem-body context behavior, use `--strict-theorem-bodies`, which isolates distinct bodies. Both modes explicitly check duplicate bodies skipped by upstream replay; temporary checking names enter neither the shared frontier nor the axiom set. Receipts and checkpoint identities bind the policy; ABI 6 invalidates old state. Strict-mode CPU and wall costs

must come from the final matrix, not historical e155 isolation runs.

Closed and some higher-order native evaluations remain unsupported; official default/fresh and operational exit conventions also differ. Editable state and arbitrary malformed metadata remain trust limits. Reports and controls are in `packaging/checker-review-*.md` and `checker/results/theorem-body-policy/`.

Two prototype interventions explain these choices. Correcting our missing persistent marking reduced eight-worker CPU by 14.13% and wall time by 5.13% in a same-executable comparison. Replacing whole-module fallback with a dependency-closed nonordinary prefix reduced a paired wall median from 257.66 to 196.67 seconds (23.67%), with CPU increasing 0.106%. These results identify engineering causes in the prototype; the release ablations test their integrated effect separately. Persistent marking occurs between joined waves. The tool exposes both disabling flags for that evaluation.

Completed segments serialize their successful prefix for reuse. Toolchain, artifacts, Lake lock and checkpoint configuration must match; a directory lock protects a live invocation. An interrupted segment remains incomplete. Receipt v1 distinguishes the accepted prefix, current-invocation CPU and previously completed segment CPU, so a cache hit is identifiable. It records Lean and dependency versions, module-set hash, accepted and rejected modules, reported axiom sets and CPU/wall clocks. A restart experiment charges restoration and restart overhead.

4.2 Audit coverage and explain trust

Audit connects source inventory, the default import graph and the actual checking receipt. `proofserve audit <project>` masks comments and strings before counting kernel-decision sites and extracts imports. Supported declarative Lake configuration gives default-root coverage; executable or unsupported configuration yields an explicit unknown unless evaluated roots are supplied. Supplied roots retain their supplied status. The receipt contributes its checked compiled closure and reported axioms; a partial receipt reports the last successful prefix. These lists enumerate axiom declarations in accepted modules, rather than minimal per-theorem dependency sets. Per-module costs require explicit source-bound observations.

The tool trusts Lean's kernel plus its artifact reader, context metadata, checked-token FFI, quiescent marking and editable local checkpoints. Recorded file identities are noncryptographic; receipt reuse is not remote attestation or a proof of checkpoint integrity. Internal symbols and serialized metadata require a matching SDK. Unknown toolchains fail clearly or explicitly select the matching official checker, identifying that path and its scope. The current receipt lacks a binding to the current source digest; audit exposes this gap. Thus acceptance, source coverage and reported axiom declarations remain separate answers.

Closure	Workers	CPU / wall	Set
Mathlib Nat.Basic	Pending	Pending	Pending
Grid	Pending	Pending	Pending
Barrier	Pending	Pending	Pending

Table 1: Release-tool scaling, awaiting the complete committed matrix. Three repeats per worker count support separate CPU and wall summaries. D labels the shared-checked default and S strict theorem-body isolation. Design implication: assess latency savings together with work inflation; prototype measurements cannot establish release performance.

5 Evaluation: performance and verification coverage

The prepared evaluation separates final performance, common-scope acceptance and audit findings; final frozen-tool results are still awaiting return. Every admitted system result must name its executable and project commits, toolchain, actual module set and complete timing boundary. Historical prototypes explain causes but cannot populate the release tables.

5.1 RQ3: original-kernel scaling and alternatives

Dependency-aware parallel replay targets elapsed time while preserving an explicit account of total CPU and additional trusted code. The release matrix uses 1, 2, 4 and 8 workers for Nat.Basic and Grid, and 1 and 8 for Barrier, with three fresh-checkpoint processes per point. Mathlib denotes the Nat.Basic closure. At eight workers, Nat.Basic and Grid each receive three no-persistence, no-split-fallback and strict-policy runs. Wall/CPU medians and ranges accompany scoped acceptance and memory observations (Table 1). The final table separates shared-checked default and any measured strict points; their CPU/wall cost comparison is reported within the same tool, input and worker configuration.

Official default and `--fresh` comparisons require matching ledgers. Lean 4.34.1 already launches module-local tasks; its default path trusts loaded imports and replays local declarations, whereas `--fresh` replays the imported final constant map sequentially [4]. ProofServe's dependency-ordered per-module ledger is a third boundary. The evaluation reports complete command costs and policy-qualified common-scope acceptance; unmatched work is identified before considering speedup ratios.

An independent kernel offers another performance/trust tradeoff. The earlier report completes

416 Comparator configurations in 80.1 active hours [10]. A paired sweep finds median nanoda/Lean time 0.21, using four nanoda threads against one Lean thread, excluding export. The pinned nanoda setup, including an enlarged checker stack, accepts 401 of 416 configurations; 15 encounter operational failures such as stack exhaustion or capacity limits, with later revisions recovering most failing cases. Nanoda is a separately implemented kernel [9]; ProofServe instead uses Lean's original kernel plus coordination. The report's result motivates this alternative, without establishing which route is faster under our evaluation conditions.

Library-checking coverage. A stopped library check reports how much was verified. The bounded attempt has not returned a final scope count. The report records unique accepts against the input frame, attempted roots, proof rejections, operational stops and unattempted scope, with tool/input commits and available official references. A verified prefix establishes that scope; unfinished work has no verdict or whole-library parity claim.

5.2 RQ4: default targets miss a third of decision sites

The full-frame import graph misses 19.78% of source files and 33.30% of lexical kernel-decision sites at our corpus pin. It covers 130,834 of 163,097 source units (including configuration), leaving 32,263 outside; the omitted files contain 451,008 of 1,354,379 sites. The earlier report finds 30,697 files outside its default graph, about 25% of its 122,459-node source inventory including the root [10]. The revisions and root-count conventions differ, but both demonstrate that a successful default build covers a subset of the source tree. The 33.30% figure describes decision sites, not module count.

Lexical density makes the workload difference visible even before timing. After masking comments and strings, the AI corpus contains 1,354,379 kernel-decision sites in 5,848 files; the pinned Mathlib subtree has seven sites in four of 8,529 files. Normalizing by decimal source megabytes gives a 10,668-fold density ratio. An audit should expose these labels alongside measured costs and uncertainty, rather than convert a site count into CPU work.

Repeated verdicts provide a second, separate kind of evidence. A historical panel checks 2,524 eligible stored-proof targets from three modules at 1, 2, 4 and 8 workers, three repeats each: 30,288 accepted checks and 22,716 one-worker/higher-worker comparisons. Verdicts, diagnostics and allocation counts agree in every observed pair, while elapsed time varies. These fixed-panel observations complement the installed-package rejection controls below.

The prepared external set pins Newspeak at `e2232d8f`, Solar at `f4f5c9a1`, and the public `samuelburbulla/proofs` collection at `3042904c`. The latter describes itself as AI-generated proofs; the model and human editing fraction are undisclosed. The final run must use an extracted frozen installation, fresh checkpoints and the original compiled inputs, with a common-root official comparison. Failed or resource-interrupted calls are operational outcomes rather than kernel rejection; incomplete whole-request CPU stays unknown.

Project	Final checking costs	Parity
Newspeak	Pending	Pending
Solar	Pending	Pending
AI proofs	Pending	Pending

Table 2: Final installed-tool external evaluation slots. CPU and wall seconds are separate; the artifact retains repeat counts and full ranges. Official default trusts imported proofs, so common local-root parity does not establish equal full-closure obligations or a speedup ratio. Design implication: retain scope, failures and input/tool commits.

Audit gives actionable coverage and axiom findings. Static audit covers Newspeak's 105 local files and Solar's 12 of 32 files in the default import graph. Solar's 20 omitted files are inputs to a separate EVM checking flow, which this evaluation did not execute. Newspeak's receipt lists a project `native_decide` axiom alongside declared closure axioms; a declaration inventory alone cannot assign those axioms to a particular theorem. Project pins, source-repair details and all failed outcomes are retained in `data/external-projects/`.

The `e000ed62` portable preview passes relocation, installed check/audit, default/strict forwarding, unsupported-version refusal and three original-kernel bad-proof controls. Three external workflows complete; their local-root sets match official default. The AI case completes after two timeouts and continuation, with whole-request CPU unknown. An offline container passes project and service controls without host mounts. Test-time packaging inputs were dirty: these are implementation-lane functional controls, not immutable release acceptance or independent runtime review. Final timings use fresh state. `serve` remains Kimina-compatible; service features are outside this evaluation.

Artifact and reproduction. The artifact is at https://github.com/yunwei37/proofserve. Use `proofserve check /path/to/project` and `proofserve audit /path/to/project`; build with `cd checker; lake build`. `docs/paper/generate_census.py` and `docs/paper/generate_system.py` consume committed CSVs. Stopped-subset generation retains unmeasured identities and unsuccessful attempts.

6 Limitations and related work

Selected cost differences do not define a universal two-class law. The dependency-ordered compilation prefix does not establish representativeness; its missing tail includes memory-intensive families. The original two-draw strata and fragile heavy-tail uncertainty cannot fill that gap. Lexical markers are proxies, not runtime frequencies. Selected closures overlap, module-local replay trusts imports, and target windows exclude shared preparation. Cross-protocol ratios, cache/order effects, hardware differences and provisional Subexponential-SAT controls limit generalization. Failed and blocked modules have no successful cost; their exclusion from cost distributions is explicit. Dynamic phase weights remain unmeasured.

Finite verdict agreement does not prove arbitrary-extension safety, future budget sufficiency or soundness of the coordinator. Reported axiom unions retain their actual scope: acceptance can depend on axioms, unsafe/partial exclusions remain visible, and semantic statement faithfulness requires separate review. Current source binding, malicious checkpoint integrity, executable Lake configuration and uncompiled/generated sources require explicit additional evidence. Release scaling and resumption costs remain incomplete. Historical external observations have one successful arm each and common-root rather than full-dependency official parity; final evaluation is pending; the AI disclosure is a collection-level attribution, not an independent authorship audit.

Parallel proof checking and asynchronous tasks have precedents in Isabelle, Coq and Lean [13, 1, 3]. Independent checkers and Lake's shipped workflow already address rechecking [6, 7]; the same-corpus tooling report also proposes standardized timing, axiom and hash receipts [10]. Kimina and AXLE supply verification services [12, 14]. Our contribution is the measured boundary between checking and recomputation and its integration into a usable original-kernel tool, rather than new parallelism, receipt or serving mechanisms.

7 Conclusion

Matched measurements show both cheap proof checking and substantial certificate recomputation. The 128,331-module observed prefix yields cost lower bounds, not a full-library answer; unfinished work and P/E boundaries stay explicit. Default coverage and repeatable verdicts answer different verification questions. ProofServe makes these boundaries explicit through check, audit and scoped receipts. Review-driven repairs strengthen the tested input and receipt boundaries; independent review and the pending final evaluation must establish the release's remaining correctness and performance claims.

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